

## **MUSCLES AFFECTED**

**hip adductors, sartorius**



## **Calf**

### **22 WALL STRETCH**

1. Stand facing the wall with feet shoulder-width apart and toes approximately 2 feet (0.6 m) from the wall.
2. Lean forward, placing the hands on the wall.
3. Step back approximately 2 feet (0.6 m) with the stretch leg while flexing the opposite knee.
4. Extend the knee of the stretch leg and lower the heel to the floor to apply the stretch.
5. Repeat on the other side.

## **MUSCLES AFFECTED**

**gastrocnemius, soleus, Achilles tendon**



## 23 STEP STRETCH

1. Place the ball of one foot on the edge of a step or board 3 to 4 inches (8-10 cm) high, with the other foot flat on the step.
2. With straight legs, lower the heel of the foot on the edge of the step as far as possible.
3. Repeat on the other side.

*Note:* To stretch the Achilles tendon, complete the same stretch with 10 degrees of knee flexion.

### **MUSCLES AFFECTED**

**gastrocnemius, soleus, Achilles tendon**



## DYNAMIC STRETCHING TECHNIQUES

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## Guidelines and Precautions for Dynamic Stretching

- Carry out 5 to 10 repetitions for each movement, either in place or over a given distance.
- Where possible, progressively increase the ROM on each repetition.
- Where appropriate, increase the speed of motion in subsequent sets, but always maintain control of the motion.
- Actively control muscular actions as you move through the ROM.
- Where appropriate, try to replicate the movements required for sport performance.
- Move progressively through the ROM.
- Move deliberately through the motion but without bouncing (movement should be controlled at all times).
- Do not forsake good technique for additional ROM.

## 24 ARM SWINGS

1. Stand erect and raise the arms in front of the body until they are parallel to the floor.
2. While walking over a prescribed distance, swing the arms in unison to the right so the left arm is in front of the chest, the fingers of the left hand are pointing directly lateral to the left shoulder, and the right arm is behind the body.

3. Immediately reverse the direction of movement to swing the arms in unison to the left.
4. Movement should occur only at the shoulder joints (i.e., keep torso and head facing forward).
5. Alternate swinging the arms in unison to the right and left.

### **MUSCLES AFFECTED**

**latissimus dorsi, teres major, anterior and posterior deltoids, pectoralis major**

### **COMMON ERROR**

**Losing postural integrity as the arms swing**



**25**

**GLUTE BRIDGE**

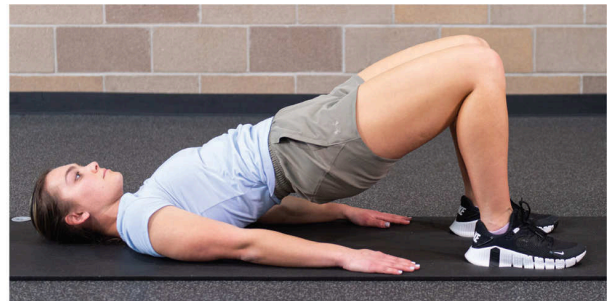
1. Lie supine with the knees flexed to approximately 90 degrees and the feet hip-width apart and flat on the floor.
2. Pushing the feet into the floor and squeezing the glutes, lift the hips up toward the ceiling until the body attains a straight-line posture.
3. Hold for a second and then return the hips to the floor.
4. The action can use a single leg as an alternative to a double leg.

### **MUSCLES AFFECTED**

**gluteus maximus, hamstrings, erector spinae**

### **COMMON ERROR**

**Using an excessive range of motion and hyperextending the lower back**



## **26 INCHWORM**

1. Stand erect with the feet placed shoulder-width apart.
2. While slightly flexing the knees, flex forward at the waist and place hands shoulder-width apart flat on the floor.
3. The weight of the body should be shifted back (i.e., not directly over the hands), with the buttocks high in the air—imagine making an inverted V with the body.
4. Move the hands alternately forward, as if taking short steps with the hands, until the body is in a push-up position.

5. Walk the legs to the hands using small steps while keeping the knees slightly flexed.
6. Repeat the motion over a prescribed distance.

### **MUSCLES AFFECTED**

**erector spinae, gastrocnemius, gluteus maximus, hamstrings, soleus, anterior tibialis**

### **COMMON ERRORS**

- **Rushing the exercise**
- **Failing to maintain postural integrity through the movement**



## **27**

## **QUADRUPED REACH AND ROTATE**

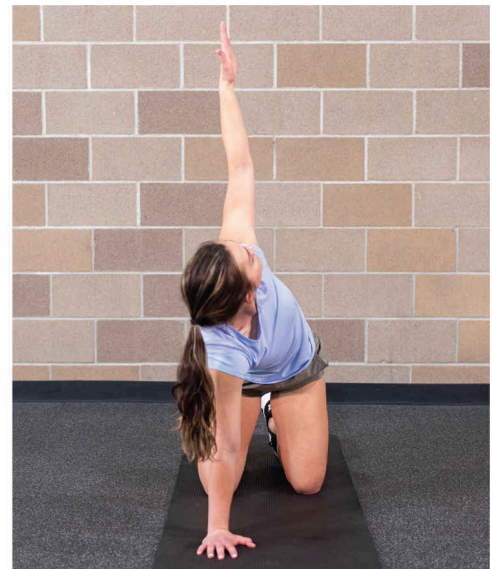
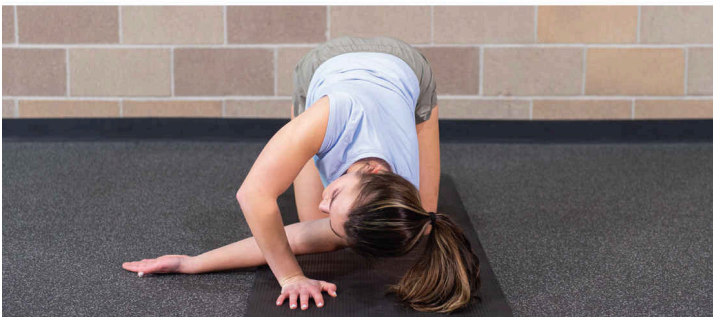
1. Assume a quadruped position.
2. Raise the left hand and extend the hand under the body, reaching as far to the right as possible.
3. Reverse the action, pulling the left hand back to the beginning position before pulling the arm back and turning the body away from the support arm.
4. Keep the lumbar spine in a neutral position so that the action occurs mainly in the thoracic spine.
5. Carry out the action in a slow and controlled manner.
6. Repeat on the other side.

### **MUSCLES AFFECTED**

**obliques, erector spinae, rectus abdominis, levator scapulae, subscapularis, rhomboids, serratus anterior, subclavius, pectoralis major, pectoralis minor, latissimus dorsi, medial deltoid**

### **COMMON ERROR**

**Losing lumbar alignment to gain more range of motion**



## **28 LUNGE WALK**

1. Stand erect with the feet parallel to each other and shoulder-width apart.
2. Take an exaggerated step directly forward with the left leg, planting the left foot flat on the floor pointing straight ahead.
3. Allow the left hip and knee to slowly flex, keeping the left knee directly over the left foot.
4. Slightly flex the right knee and lower it until it is 1 to 2 inches (3-5 cm) above the floor; the right foot should be pointed straight ahead.

5. Balance the weight evenly between the ball of the right foot and the entire left foot.
6. Keep the torso perpendicular to the floor by sitting back on the right leg.
7. Forcefully push off of the floor by extending the left hip and knee.
8. Pick up the right foot and place it next to the left foot; do not stutter-step forward.
9. Stand erect, pause, and then step forward with the right leg, progressing forward with each step.

### **MUSCLES AFFECTED**

**gluteus maximus, hamstrings, iliopsoas, quadriceps**

### **COMMON ERROR**

**Not achieving the required position at the ankle, knee, hip, and torso**



## **29 LATERAL LUNGE**

1. Stand with the feet hip-width apart and feet facing forward.

2. Take a relatively large lateral step with the right foot while keeping the left foot in contact with the floor.
3. Hinge at the hips, pushing the hips back, and flex the left hip and knee, lowering the body toward the ground until the left hip and knee are at approximately 90 degrees. The right knee remains fully extended as the left knee and hip flex.
4. Keep the body aligned within the base of support, and do not lean forward.
5. Return to the beginning position and then either repeat for the desired number of repetitions on the same leg or alternate legs for the desired number of repetitions.

### **MUSCLES AFFECTED**

**quadriceps, hip adductors, hip abductors, gluteus maximus, gluteus medius**

### **COMMON ERRORS**

- **Taking a step that is too small or too long**
- **Losing the feet alignment**
- **Not maintaining postural alignment**



## 30 LUNGE WALK WITH FRONT LEG EXTENSION

1. Take an exaggerated step directly forward with the right leg and assume a lunge stance, keeping the right knee directly over the right foot and the rear (left) knee about an inch or so off the floor (*note*: this preparatory position is not shown).
2. Flex the torso forward with the hands reaching toward the right foot. From this position, lift the hips up and back, extending the right knee to a fully (or nearly so) extended position. This will cause the left knee to extend as well and result in a stretch of the hamstring muscles.
3. After the stretch is held, extend the torso to an erect position, step the rear (left) foot forward, and repeat on the other side.

### MUSCLES AFFECTED

**hamstrings, gluteus maximus, iliopsoas, quadriceps**

### COMMON ERROR

**Losing postural integrity during the movement**



## 31 SPEED SKATER LUNGE

1. Assume an upright stance with feet hip-width apart and arms to the side (*note: this preparatory position is not shown*).
2. Shift the body weight to the left foot and at the same time drop the hips back and down, reaching the right leg behind the body and as far as possible to the opposite (right) side. The result is a fully extended rear (right) knee and a contralateral (opposite) lateral reach of the other (left) arm.
3. Push back to the beginning position and repeat on the other side.
4. For a more advanced exercise, the movement can be initiated by a lateral hop.

### **MUSCLES AFFECTED**

**quadriceps, gluteus maximus, gluteus medius, hip adductors, hamstrings**

### **COMMON ERRORS**

- **Taking a step that is too small or too long**
- **Losing postural integrity during the movement**



## **32** **LUNGE WITH OVERHEAD SIDE REACH**

1. Stand erect with the feet parallel to each other and shoulder-width apart.
2. Take an exaggerated step directly forward with the left leg, planting the left foot flat on the floor pointing straight ahead.
3. Allow the left hip and knee to slowly flex, keeping the left knee directly over the left foot.
4. Slightly flex the right knee and lower it until it is 1 to 2 inches (3-5 cm) above the floor; the right foot should be pointed straight ahead.
5. Reach up high with the right arm and bend the torso laterally toward the left leg.
6. Return to an erect torso position, and then forcefully push off the floor by extending the left hip and knee.
7. Pick up the right foot and place it next to the left foot; do not stutter-step forward.
8. Stand erect, pause, and then step forward with the right leg, progressing forward with each step.

### **MUSCLES AFFECTED**

**gluteus maximus, hamstrings, iliopsoas, latissimus dorsi, internal and external oblique, rectus femoris**

### **COMMON ERROR**

**Reaching too early before the lunge position has been attained**



## **33** **WALKING KNEE LIFT**

1. Stand erect with the feet parallel to each other and shoulder-width apart.
2. Step forward with the left leg and flex the right hip and knee to move the right thigh upward toward the chest.
3. Grasp the front of the right knee or upper shin and use the arms to pull the right knee up further and to squeeze the thigh against the chest.
4. Dorsiflex the left foot as the right hip and knee are flexed.
5. Keeping the torso erect, pause for a moment, then proceed to step down with the right leg.
6. Shift the body weight to the right leg and repeat the motion with the left leg.
7. Progress forward with each step, increasing the ROM and speed on subsequent steps.

### **MUSCLES AFFECTED**

**gluteus maximus, iliopsoas, quadriceps, hamstrings**

### **COMMON ERROR**

**Leaning the torso backward to achieve ROM**



## **34 FORWARD LUNGE WITH ELBOW TO INSTEP**

1. Stand erect with the feet parallel to each other and shoulder-width apart.
2. Take an exaggerated step directly forward with the left leg, planting the left foot flat on the floor pointing straight ahead.
3. Allow the left hip and knee to slowly flex, keeping the left knee directly over the left foot.

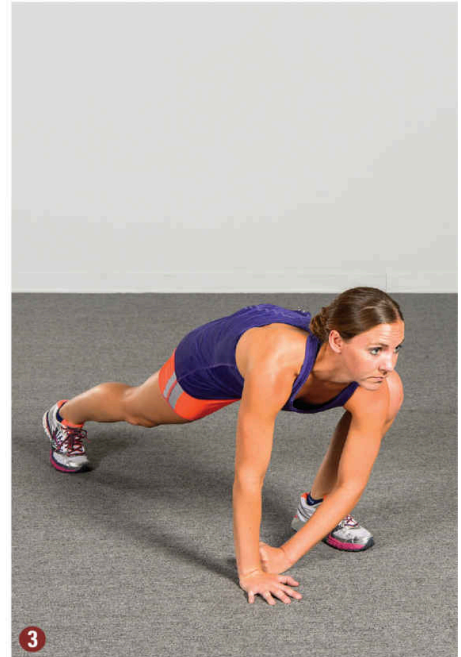
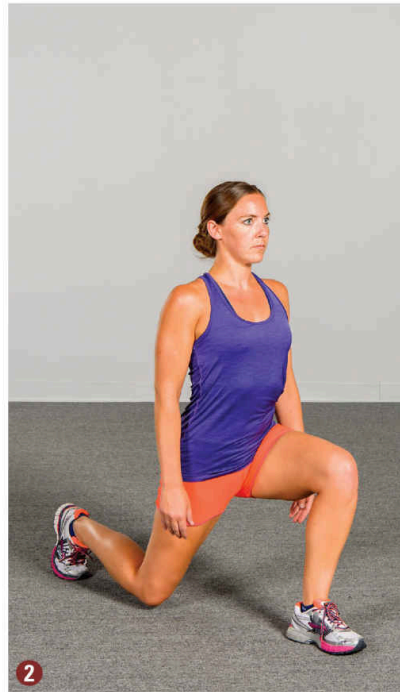
4. Slightly flex the right knee and lower it until it is 1 to 2 inches (3-5 cm) above the floor; the right foot should be pointed straight ahead.
5. Lean forward, bringing the left arm forward and touching the left elbow to the instep of the left foot; the right hand may be placed on the floor to maintain balance.
6. Lean back to return to an erect torso position, and then forcefully push off the floor by extending the left hip and knee.
7. Pick up the right foot and place it next to the left foot; do not stutter-step forward.
8. Stand erect, pause, and then step forward with the right leg, progressing forward with each step.

#### **MUSCLES AFFECTED**

**biceps femoris, erector spinae, gastrocnemius, gluteus maximus, hamstrings, iliopsoas, latissimus dorsi, internal and external oblique, quadriceps, soleus**

#### **COMMON ERROR**

**Losing postural integrity in an attempt to achieve greater ROM**



## 35 HEEL-TO-TOE WALK

1. Stand erect with the feet parallel to each other and shoulder-width apart.
2. Take a small step forward with the right leg; place the heel of the right foot on the ground first and then continue to dorsiflex the foot.
3. Immediately roll forward and rise up as high as possible onto the ball of the right foot.
4. Swing the left leg forward in order to take another small step.
5. Repeat with the left leg, progressing forward with each step.

### MUSCLES AFFECTED

**gastrocnemius, soleus, anterior tibialis**

### COMMON ERROR

## Failing to maintain a vertical posture and losing balance



### 36 WALKING OVER AND UNDER

1. Stand erect with the feet parallel to each other and shoulder-width apart.
2. Flex the left hip and knee, and then abduct the left thigh until it is parallel to the floor.
3. Step laterally to the left, stepping laterally over the first hurdle.
4. Place the left foot firmly on the ground, shift the body weight to the left leg, and then proceed to lift the right leg over the first hurdle.
5. After lifting the right leg over the hurdle and placing the right foot firmly on the ground, stand erect, pause, and then flex the hips and knees and dorsiflex the ankles to assume a full squat position.
6. Extend the left leg laterally, as if performing a lateral lunge.
7. Keeping the body weight low, move the body laterally, ducking under the second hurdle.

8. Stand erect, pause, and repeat the motion in the opposite direction, ducking under the second hurdle and stepping over the first.

*Note:* If hurdles are not available, athletes can perform the exercise by mirroring the motion of trying to step laterally and then under a hurdle.

### **MUSCLES AFFECTED**

**hip abductors, hip adductors, gastrocnemius, gluteus maximus, hamstrings, iliopsoas, rectus femoris, soleus**

### **COMMON ERRORS**

- **Leaning the torso backward to achieve ROM in the over section**
- **Flexing the torso forward rather than squatting to achieve ROM in the under section**



## 37 INVERTED HAMSTRING STRETCH

1. Stand tall and take a small step forward with the left leg.
2. Flex at the waist.
3. Reach forward with the left arm and simultaneously reach back with the right leg.
4. Try to keep the hips square.
5. Reach a position in which you feel a stretch in the hamstrings but eventually aim to attain a position in which the body is parallel to the floor.
6. Reach the right hand down toward the floor.

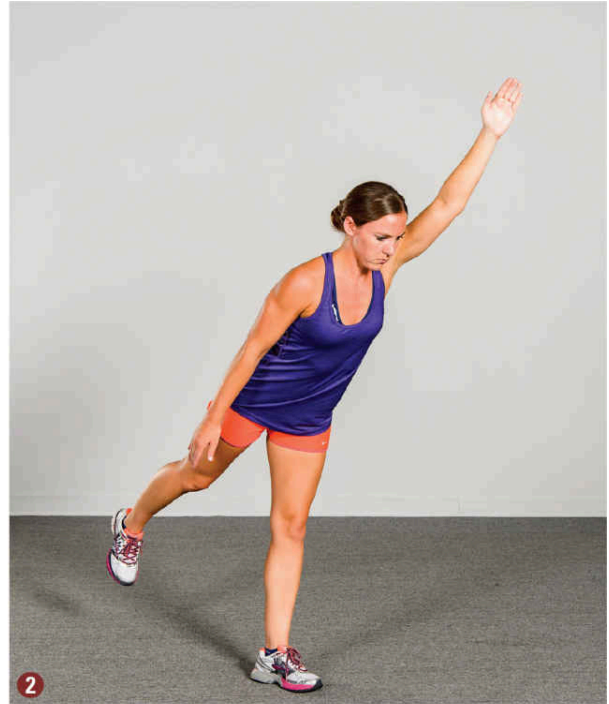
7. Return to the start by actively using the hamstrings and gluteals of the planted leg.
8. Take a step forward and repeat on the other side.

#### **MUSCLES AFFECTED**

**gluteus maximus, hamstrings, hip abductors, hip adductors, erector spinae**

#### **COMMON ERROR**

**Performing the exercise too quickly and losing postural integrity**



## 38 STRAIGHT-LEG MARCH

1. Assume a tall standing position with both arms outstretched in front of the chest.
2. Rise up onto the toes of the left leg and simultaneously, using a straight-leg action, raise the right leg forward and up, aiming to actively move it through as large a ROM as possible.
3. Once the highest point is reached, actively pull the leg back to the beginning position.

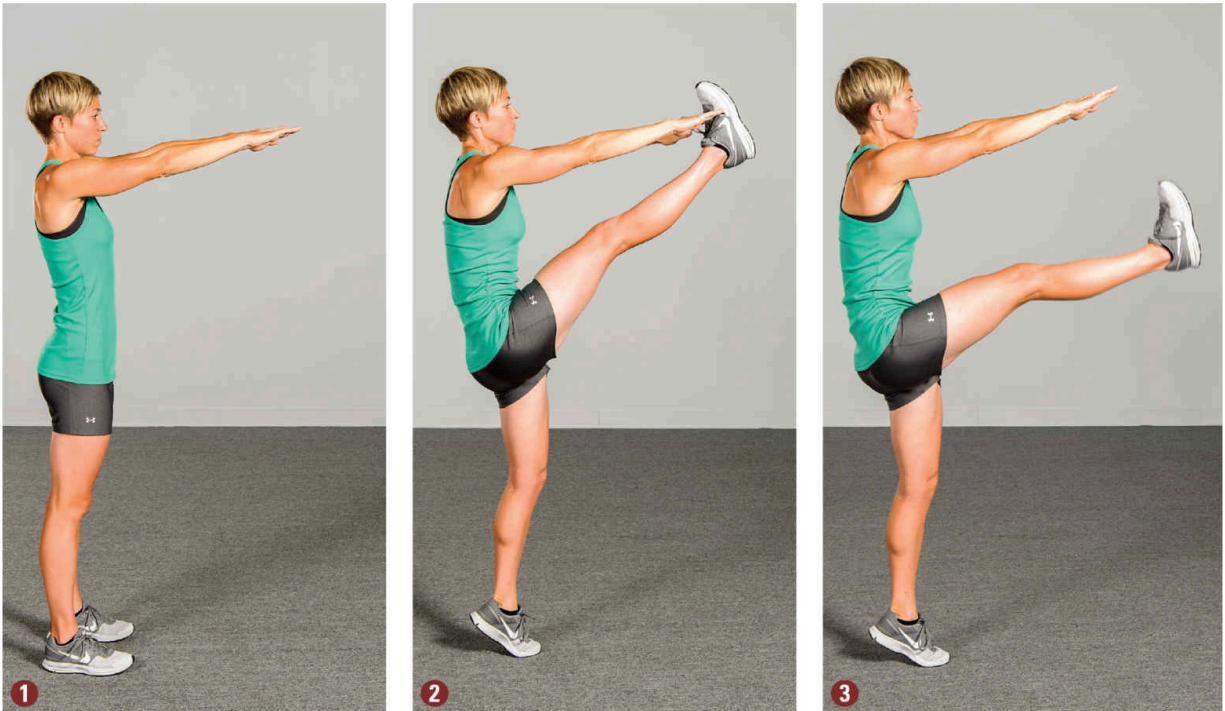
4. Maintain the tall standing posture throughout.
5. Repeat on alternate legs while moving forward.

### **MUSCLES AFFECTED**

**gluteus maximus, hamstrings, iliopsoas, rectus femoris**

### **COMMON ERROR**

**Using excessive momentum to increase the ROM**



**39**

## **BEAR CRAWL**

1. Assume a push-up position.
2. Reach forward with the right arm and the left leg, moving the body forward in a crawling motion.
3. Reach forward with the opposite leg and hand combination.
4. Keep the knees 1 to 2 inches (3-5 cm) off the ground.
5. Repeat in a rhythmic manner for the desired distance or repetitions.

6. As competency increases, the ROM can increase. Additionally, the crawl can be performed backward.

### **MUSCLES AFFECTED**

**erector spinae, gastrocnemius, gluteus maximus, deltoids, iliopsoas, latissimus dorsi, internal and external oblique, quadriceps**

### **COMMON ERRORS**

- **Positioning the hips too high**
- **Moving the back out of a neutral position**



**40**

## **SPIDERMAN CRAWL**

1. Assume a push-up position but with elbows flexed so that the position is lower than normal.
2. Lift and externally rotate the left leg, bringing the knee forward at an angle outside the left elbow.
3. Walk the hands forward and across the body.
4. As this progresses, bring the right leg forward outside the right hand and repeat for the given distance.
5. Repeat, alternating sides, for the required distance.

### **MUSCLES AFFECTED**

**biceps femoris, erector spinae, gastrocnemius, gluteus maximus, hamstrings, iliopsoas, latissimus dorsi, internal and external oblique, quadriceps, soleus**

### **COMMON ERRORS**

- **Positioning the hips too high**
- **Moving the back out of a neutral position**



## **PNF STRETCHING TECHNIQUES**

- 41** Calves [408](#)
- 42** Chest [408](#)
- 43** Groin [408](#)
- 44** Quadriceps and Hip Flexors [409](#)
- 45** Shoulders [409](#)

## **Guidelines and Precautions for PNF Stretching**

- Decide on the muscle to be stretched, the ROM needed for the activity, and whether PNF methods are best suited.
- Decide on the stretch to be performed and the type of PNF stretching best suited to the situation.
- Communicate clearly with your partner before and throughout the stretch.
- Contraction durations should be of a moderate intensity.
- Contractions are typically sustained for 6 to 10 seconds.
- Passive stretch durations are typically 30 seconds.

- Rest periods of 30 seconds are recommended between repetitions, and 2 to 4 repetitions are recommended per stretch.
- Ensure a thorough warm-up is performed prior to PNF stretching.
- Use moderate contraction intensities rather than high intensities, especially if stretching smaller joints.
- Progress the ROM gradually during the protocols.
- Effective communication with your partner is essential to ensure the exercise is performed safely.
- Take special care if stretching smaller joints such as the shoulders.

## 41 CALVES



## 42 CHEST



**43**

**GROIN**



**44**

**QUADRICEPS AND HIP FLEXORS**



**45**

## SHOULDERS



### KEY TERMS

active stretch  
agonist contraction  
autogenic inhibition  
ball-and-socket joint  
ballistic stretch

central nervous system (CNS) preparatory exercise  
contract-relax  
dynamic flexibility  
dynamic stretch  
elasticity  
ellipsoidal joint  
fibrosis  
flexibility  
general warm-up  
Golgi tendon organ (GTO)  
hinge joint  
hold-relax  
hold-relax with agonist contraction  
mobility  
mobility drills  
muscle spindles  
musculotendinous unit (MTU)  
passive stretch  
plasticity  
postactivation performance enhancement (PAPE)  
potentiation  
proprioceptive neuromuscular facilitation (PNF)  
raise, activate and mobilize, and potentiate (RAMP)  
range of motion (ROM)  
reciprocal inhibition  
specific warm-up  
static flexibility  
static stretch  
stretch reflex

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## **STUDY QUESTIONS**

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1. Which of the following is not a temperature-related effect of a warm-up?
  - a. an increase in core body temperature

- b. the disruption of transient connective tissue bonds
  - c. increased blood flow to working muscles
  - d. enhanced neural function
2. How does the activate and mobilize phase of the RAMP protocol differ from traditional static stretching?
- a. It emphasizes holding positions to lengthen muscles and improve flexibility.
  - b. It focuses on developing mobility through dynamic movements that build movement competency.
  - c. It eliminates movement patterns in favor of isolated joint stretches.
  - d. It is only used for recovery sessions and not before performance.
3. Which of the following statements best reflects advice for CNS stimulation?
- a. The impact of CNS stimulation is highly individual and should be considered on a cost-benefit basis.
  - b. Stimulation will always require the use of resistance training exercises.
  - c. Stimulation can only be elicited through the application of high resistances.
  - d. Beginners are likely to benefit to a greater degree than highly trained athletes.
4. Which of the following is a potential reason for an increase in ROM according to the sensory theory?
- a. autogenic inhibition
  - b. increases in stretch tolerance
  - c. a change in the mechanical properties of the musculotendinous unit
  - d. a change in muscle architecture
5. Which of the following statements about static stretching is supported by current research evidence?

- a. Total stretch time accumulated throughout the day may be as important as individual stretch duration.
- b. The duration of a single stretch is more important than how often stretching is performed.
- c. Stretching for a maximum of 10 seconds produces superior results than stretching for 30 seconds.
- d. Stretching should be done to the point of pain to increase range of motion.